

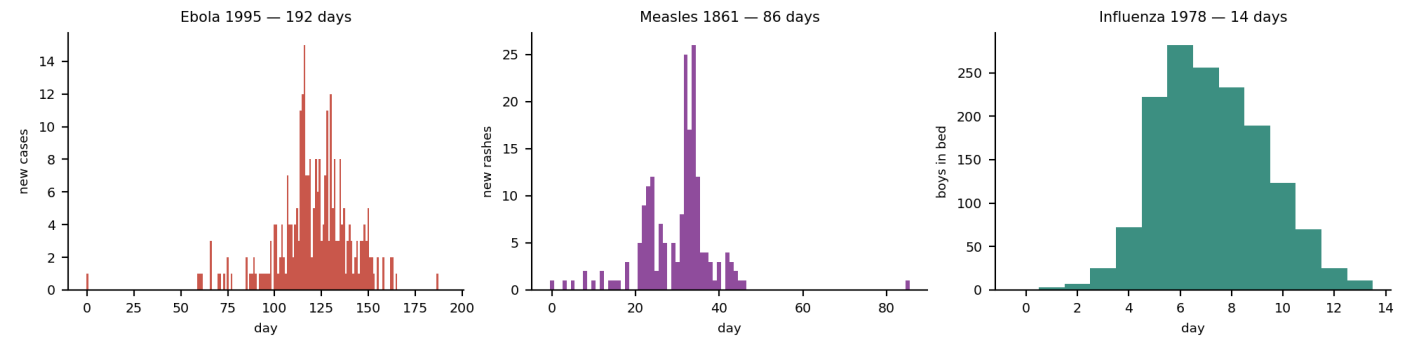
Four Outbreaks — Comparative Overview

Companion to the four dossiers · all figures computed before any model was fitted

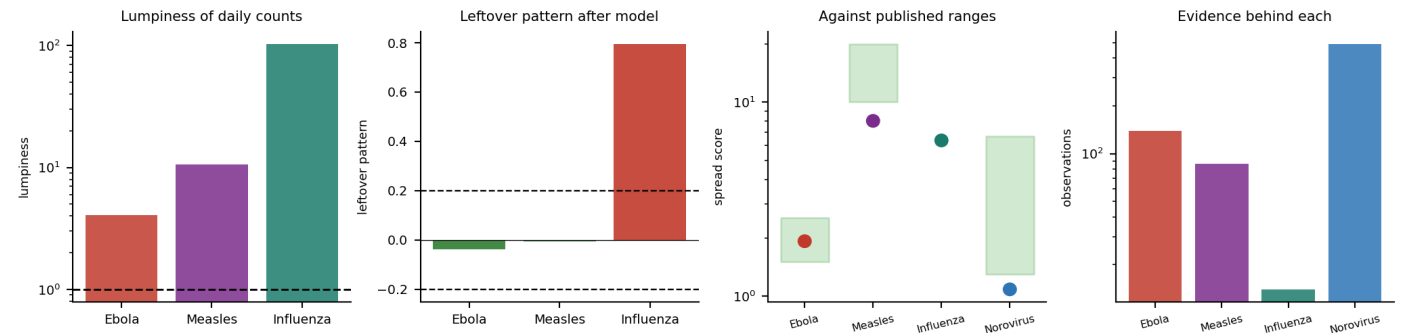
Where these numbers come from

Source	What it is	Used for
RAW records	the outbreak as originally recorded, before any decision	the evidence that justifies every decision taken
CANONICAL dataset	the frozen analysis file after those decisions	what the model actually reads

This shows that both are reported, side by side, in every dossier. A problem that has been fixed is shown as fixed with its before and after numbers — it is not deleted from the report. That trail is what lets someone check the work.



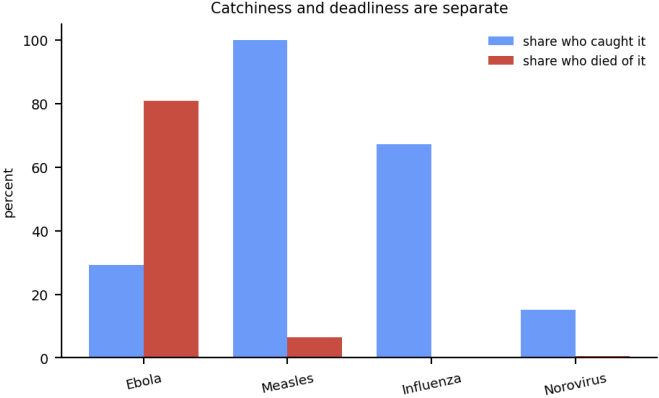
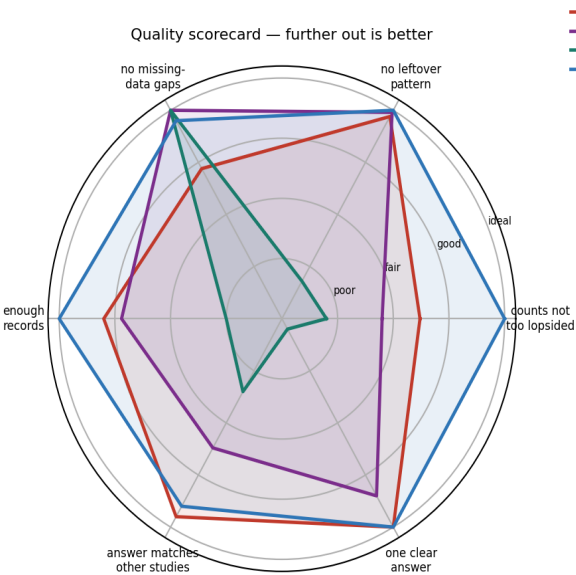
This shows that each outbreak rose once and fell once. That single-hump shape is what makes them comparable to one another at all.



This shows that the second panel is the decisive one. Green means the model explained the timing and left nothing behind; red means it did not. Only influenza is red.

Scorecard and Severity

Two views that do not need any statistics to read



This shows that further from the centre is better. Norovirus is nearly a full hexagon. Influenza collapses inward on four of six spokes — that shape is the whole argument for labelling it.

This shows that catchiness and deadliness are independent. Measles infected everyone and killed one in sixteen; Ebola infected few and killed four in five. Neither number alone says how serious an outbreak was.

Every concern raised, and where it ended up

Outbreak	Concerns raised	Fixed	Handled another way	Left open	Verdict
Ebola	6	6	0	0	CLEARED
Measles	6	4	1	1 (disclosed)	CLEARED
Norovirus	6	5	1	0	CLEARED
Influenza	6	2	1	3	LABELLED

This shows that three outbreaks came through with everything either fixed or explained. Influenza has three items that no method can fix, which is why it is presented separately.

Reference Numbers and Reading Guide

The technical form of everything above

Outbreak	Var/Mean	Skew	Zero-day share	lag-1 RAW	lag-1 AFTER model	Const-fit R0	Needs GRW
Ebola	4.06	1.95	0.41	+0.719	-0.0374	1.92	no
Measles	10.64	3.31	0.59	+0.739	-0.0047	8.01	no
Influenza	103.64	0.45	0.00	n/a	+0.7961	6.35	YES
Norovirus	n/a	n/a	n/a	n/a	n/a	1.08	n/a

This shows that the two lag-1 columns are the before and after. Ebola goes from a strong pattern to nothing; influenza stays high. That contrast drove every modelling decision in this study.

Vocabulary used in these dossiers

Plain term	Technical term	Meaning
spread score	R0, basic reproduction number	average number of people one infected person passes it to, in a group with no immunity
lumpiness	overdispersion (variance / mean)	how much more the daily counts jump around than a simple tally would predict
leftover pattern	residual autocorrelation	whether the model's errors still follow a trend after it has done its best
forgiving count	negative binomial likelihood	a way of counting that expects quiet days followed by clusters
share who fall ill	attack rate / final size	fraction of the whole group that eventually catches it
change-over-time allowance	Gaussian random walk on log beta(t)	letting contagiousness drift day by day instead of holding it fixed

This shows that six terms cover every chart and table in this set of documents.